

especially in tissues such as skeletal muscle, which helps explain why high systemic doses or conjugated delivery approaches are often needed to achieve robust pharmacodynamic effects (Heo et al., 2024).

Pharmacodynamic duration is therefore not determined by plasma half-life alone. PNA can bind targets with high affinity and is biologically stable, but in vivo duration is often constrained by delivery, systemic exposure and tissue retention (McMahon et al., 2002). Phosphorothioate ASOs often show durable pharmacodynamic effects because they accumulate in tissues and are cleared slowly from intracellular and tissue compartments; consequently, tissue PK is often more informative than plasma PK for predicting response (Geary et al., 2015). Morpholinos may have relatively short plasma exposure but can still produce sustained mechanistic effects with repeated dosing, as reflected by the weekly dosing schedules used for approved DMD PMO drugs such as eteplirsen (FDA, 2016).

These pharmacological differences also shape the translational challenges of each platform. For PNA, the major challenge is converting excellent binding affinity into reliable in vivo activity, because unconjugated PNA has poor cellular uptake and limited systemic exposure. For DNA ASOs, the challenge is to balance potency, tissue distribution, RNase H activity, protein binding, immune stimulation and toxicity within a clinically tractable chemistry design (Geary et al., 2015). For morpholinos, the principal challenge is achieving sufficient intracellular concentrations in target tissues, particularly muscle, without requiring impractically high doses or introducing toxicity through delivery-enhancing conjugates (FDA, 2016; Heo et al., 2024).

PNAs are typically synthesized using solid-phase peptide synthesis. Unlike DNA and RNA synthesis, PNA assembly involves the stepwise coupling of protected PNA monomers, each composed of a nucleobase attached to an N-(2-aminoethyl)-glycine backbone, onto a solid support. The growing oligomer is extended through amide bond formation, usually using Boc- or Fmoc-protected monomers, followed by cleavage from the resin and purification by chromatographic methods. This peptide-like synthetic route provides considerable chemical flexibility, allowing functional groups, fluorophores, cell-penetrating peptides, intercalators and other delivery motifs to be introduced during or after synthesis to improve binding, detection, solubility or uptake (Nielsen et al., 1991; Egholm et al., 1993; El-Fateh, Chatterjee and Zhao, 2024).

This synthetic flexibility has enabled the development of chemically modified and conjugated PNAs designed for specific biological and biophysical applications. Bruce Armitage and collaborators expanded the chemical and supramolecular applications of PNAs through PNA-containing

hybrids, dye-PNA assemblies, strand invasion and quadruplex-targeting PNAs. For example, Smith, Olson and Armitage demonstrated that PNA-containing hybrids could template the spontaneous assembly of helical cyanine dye aggregates, highlighting the use of synthetic PNAs as programmable scaffolds for molecular recognition. Later work further developed PNAs as tools for targeting structured nucleic acids, including G-quadruplexes and mixed-sequence duplex DNA (Datta and Armitage, 2001; Datta, Schmitt and Armitage, 2003; Rapireddy et al., 2007). Together, these studies show the versatility of PNA synthesis for broader chemical biology toolkits.

Although PNAs have long been viewed as attractive therapeutic molecules, their clinical translation has been slow largely because unmodified PNAs have poor cellular uptake and require chemical modification or delivery strategies to reach intracellular targets. The most advanced example is OLP-1002, an OliPass PNA-derived antisense compound targeting SCN9A/Nav1.7 for osteoarthritis pain; a Phase 2a trial evaluating subcutaneous OLP-1002 enrolled 134 participants and was completed in 2023 (ClinicalTrials.gov, 2023).

OLP-1002, an OliPass PNA-derived antisense compound designed to inhibit SCN9A/Nav1.7, was evaluated in a two-stage Phase 2a trial for moderate-to-severe osteoarthritis pain in the knee and/or hip. The study enrolled 134 participants and was completed in mid-2023, with no formal results yet posted on ClinicalTrials.gov. Although an early interim analysis suggested a potential analgesic signal for the 2µg dose, with greater pain reduction than placebo at Day 43, the final sponsor-reported outcome did not confirm efficacy. In the expanded placebo-controlled stage, placebo responses exceeded those seen with OLP-1002: OliPass reported mean six-week pain reductions of 41% for placebo compared with 22% and 25% for the 1 µg and 2 µg OLP-1002 groups, respectively, and stated that OLP-1002 was “inferior to placebo” in this Phase 2a trial (OliPass Corporation, 2023b).

Another example is EOM613, described in its trial record as a peptide nucleic acid with immune-modulating properties, which entered a proof-of-concept study for severe COVID-19, although the trial status remains unclear (ClinicalTrials.gov). EOM613 is being developed by EOM Pharmaceuticals and appears to be the same historical product family previously known as Product R, OHR118, AVR118 or OHR/AVR118. Unlike a conventional single-molecule antisense PNA drug, EOM613 is best understood as a complex peptide–nucleotide solution containing peptide and nucleoside/nucleotide components, including a peptide–oligonucleotide conjugate. Its proposed pharmacology is “dynamic dual-action” immunomodulation: it is reported to regulate monocyte/macrophage cytokine responses, with both anti-inflammatory and pro-inflammatory

effects depending on immune-cell activation state. Early human data include open-label rheumatoid arthritis and cancer-cachexia studies, and an open-label Phase 1/2a COVID-19 study in hospitalized patients; these studies generally suggested tolerability and possible clinical or biomarker activity, but most efficacy evidence remains exploratory, small, open-label or sponsor-reported. In 2026, EOM reported FDA clearance for a Phase 2a cancer-cachexia study and plans for an additional study on Crohn's disease.

Therapeutic Modalities and Mechanisms

RNA sits at the interface of genetic information and biological function, enabling control over protein expression without permanently altering the genome. Despite its transient nature, targeting RNA can produce robust and often long-lasting clinical effects, making it a versatile platform for a wide spectrum of therapies.

One of the major advantages of RNA therapeutics is their ability to precisely and programmatically regulate gene expression. Depending on the therapeutic objective, these molecules can either enhance protein production through mechanisms such as microRNA inhibition and correction of RNA splicing or suppress the expression of harmful genes using strategies including RNase H-mediated transcript degradation and siRNA-directed RNA cleavage. Unlike many conventional small-molecule drugs, which often rely on broad chemical interactions with proteins, RNA therapeutics function through predictable sequence complementarity, enabling highly selective targeting of specific transcripts. This sequence-driven precision not only improves specificity but also supports a more rational drug discovery process, where candidate therapies can be identified and optimized through precision-based genetic and molecular screening approaches.

Beyond gene silencing and activation, RNA therapeutics can reprogram RNA processing and function. Technologies such as exon skipping, cryptic exon modulation, and isoform switching can alter the proteins produced by cells without permanently changing the genome itself. More advanced strategies employ enzymes such as Adenosine Deaminase Acting on RNA (ADAR) to perform site-specific RNA base editing, enabling transient and potentially reversible correction of pathogenic mutations. Collectively, these approaches highlight how RNA therapeutics move medicine closer to a rational and modular framework for drug development, where treatments are engineered based on precise molecular knowledge of disease pathways rather than discovered primarily through empirical screening.

The first ASO clinically approved



Fomivirsen (Vitravene™) was the first gene-silencing antisense therapy approved by the FDA, receiving approval in 1998. Originally discovered by a group at the NIH, it was first licensed by Isis Pharmaceuticals and later sublicensed to Novartis. The compound consists

of a simple 21-base, DNA oligonucleotide in which a sulfur atom replaces one of the non-bridging oxygen atoms in the phosphate backbone, forming phosphorothioate linkages between nucleosides. Vitravene was developed to treat cytomegalovirus (CMV) retinitis in immunocompromised patients and functions by binding to mRNA transcripts that encode IE2 proteins, essential for viral replication (Geary, Henry, and Grillone, 2002).

Preclinical and in vitro studies demonstrated potent antiviral activity, with mean 50% inhibitory concentrations (IC₅₀) ranging from 0.03 to 0.2 µM across multiple cell lines and a median effective concentration (EC₅₀) of approximately 0.34 ± 0.25 µM in human fibroblasts (Azad et al., 1993; Anderson et al., 1996).

Fomivirsen (ASO)	0.03–0.3 µM	Very potent, sequence-specific
Ganciclovir	1–10 µM	Antiviral comparator
Ibuprofen	1–20 µM	Moderate potency
Aspirin	10–1000 µM	Low potency
Paracetamol	100–1000 µM	Very low potency
Imatinib	~0.3 µM	High potency targeted drug

Vitravene demonstrated striking potency with EC₅₀ values in the range of 0.03–0.3 µM, highlighting the robustness of a sequence-specific mechanism of action. In comparison, traditional antivirals like ganciclovir showed lower potency (EC₅₀ ≈ 1–10 µM), while commonly used small molecules such as ibuprofen (1–20 µM) and aspirin (10–300 µM) exhibited only moderate to low activity. Paracetamol was even less potent (100–1000 µM), underscoring the substantial gap between conventional drugs and targeted nucleic acid therapeutics. Thus, reinforcing the promise of rational, sequence-driven drug design. Despite its scientific significance, it was ultimately a commercial failure. The drug was withdrawn in 2000, largely due to the success of highly effective antiretroviral therapies for HIV, which dramatically reduced the incidence of CMV retinitis, the very condition Vitravene was designed to treat.

The clinical landscape of RNA therapeutics has expanded substantially, encompassing a diverse and increasingly sophisticated set of approaches that act across multiple layers of gene regulation and cellular function. Established commercial therapies include RNase H–mediated knockdown (e.g., inotersen for transthyretin amyloidosis), siRNA-driven

RISC-mediated degradation (e.g., patisiran and inclisiran), aptamers that bind and inhibit protein targets (e.g., pegaptanib/Macugen), RNA splicing modulation (e.g., nusinersen), and mRNA-based strategies for both protein replacement and vaccination (e.g., mRNA COVID-19 vaccines) (Adams et al., 2018; Ray et al., 2017; Hoy, 2018; Finkel et al., 2017; Polack et al., 2020).

Beyond these established modalities, additional and conceptually distinct mechanisms of action are emerging. Among them, microRNA (miRNA)-based therapies enable network-level regulation by modulating entire gene programs either through inhibition (anti-miRs; e.g., miravirsen targeting miR-122 in HCV) or restoration (miRNA mimics; e.g., MRX34 in oncology) rather than acting on single transcripts (van der Ree et al., 2017; Beg et al., 2017). In parallel, RNA activation (RNAa) has emerged as a clinically validated strategy for upregulating endogenous gene expression using small activating RNAs (saRNAs).

Complementing activation-based approaches, RNA editing therapeutics particularly those leveraging adenosine-deaminases-acting-on-RNA (ADARs) introduce the ability to precisely modify RNA sequences at the transcript level. These systems enable site-specific A-to-I base conversion, which is interpreted as guanosine during translation, allowing correction of pathogenic mutations without permanent genomic alteration. Early clinical programs, such as WVE-006 for alpha-1 antitrypsin deficiency, demonstrate the translational potential of this approach by recruiting endogenous ADAR enzymes to restore functional protein expression in patients (Rainaldi, Mali and Nourredine, 2025).

Beyond editing and activation, additional emerging modalities further broaden the mechanistic scope of RNA therapeutics. Non-degradative antisense oligonucleotides can function through steric blocking to inhibit translation or disrupt RNA–protein interactions without inducing RNA cleavage, as seen in splice-switching therapies. RNA molecules can also act as innate immune agonists, for example synthetic RNAs that activate TLR7/8 or RIG-I pathways in oncology. Advances in expression platforms, including self-amplifying RNA (e.g., saRNA vaccine candidates such as ARCT-021) and circular RNA constructs, enhance the magnitude and durability of protein production (Blakney et al., 2021). Finally, RNA decoys and scaffolds can sequester proteins or regulatory RNAs, expanding RNA therapeutics into modulation of protein–RNA interaction networks.

These developments reflect a clear shift from first-generation paradigms centered on gene silencing or replacement toward a more expansive framework in which RNA therapeutics can activate, edit, regulate, and reprogram cellular behavior with increasing precision. Clinical examples spanning RNAi, antisense, RNAa, miRNA modulation, and RNA

editing underscore that this mechanistic diversity is already being realized in the clinic, marking a transition toward a more versatile and programmable class of medicines (Adams et al., 2018; Reebye et al., 2018; Rainaldi, Mali and Nourreddine, 2025).

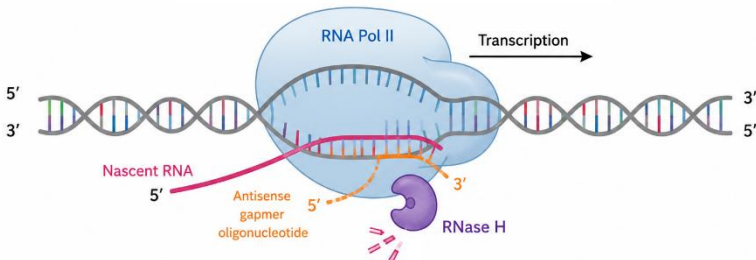
RNase H–Mediated Gene Silencing

“Die Nucleinsubstanzen liefern bei ihrer Spaltung verschiedene stickstoffhaltige Spaltungsprodukte, welche zu den wichtigsten Bestandteilen des Zellkerns gehören.”

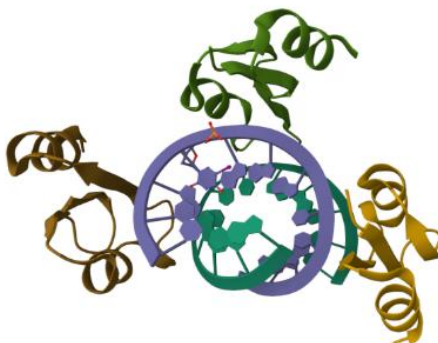
--Kossel, A., 1910. Die chemische Zusammensetzung der Zellkerne

Albrecht Kossel working at the University of Heidelberg first described nucleic substances in terms of their “cleavage products” (*Spaltungsprodukte*), illustrating that nucleic acids could be broken down into distinct nitrogen-containing bases. For this work Dr. Kossel was awarded the Nobel Prize in Physiology or Medicine. Kossel’s observations laid the foundation for the discovery of nucleases; enzymes specialized in the cleavage and degradation of DNA and RNA. During the mid-twentieth century, the identification of bacterial restriction endonucleases transformed molecular biology by enabling precise DNA manipulation for cloning, sequencing and recombinant DNA technology (Arber, 1979). Since then, nucleases including TREX1, EXO1 and MRE11 have been recognized as essential regulators of DNA replication, repair, apoptosis and innate immune defense in human systems, while also becoming indispensable tools in biotechnology, genome engineering and CRISPR-based applications (Roberts, 2005; Mazur and Perrino, 2022).

In 1976, venture capitalist Robert Swanson and biochemist Dr. Herbert Boyer founded Genentech to commercialize recombinant DNA methods pioneered by Boyer and Stanley Cohen, demonstrating that human genes could be inserted into bacteria to produce therapeutic proteins. This work led to the first successful production of human proteins in bacteria, including synthetic human insulin, which became the first recombinant pharmaceutical approved for clinical use in 1982. Genentech had established a new model for advancing therapeutics; one that deeply integrated our understanding of biology with our ability to use it for human therapeutics; a field we call Biotechnology.



RNase H is an endonuclease that plays a fundamental role in maintaining genome stability by selectively degrading the RNA strand of RNA–DNA hybrids. During DNA replication, it removes RNA primers required for initiation, ensuring their replacement with DNA and preserving genomic integrity. It also contributes to DNA repair by removing ribonucleotides, preventing structural instability and mutations (Eder, Walder and Walder, 1993). In addition, RNase H resolves R-loops that form during transcription, which, if accumulated, can interfere with replication and lead to DNA damage. Collectively, these functions establish RNase H as a key enzyme linking nucleic acid metabolism with genomic integrity (White, R.L. and Davis, R.W., 1976).



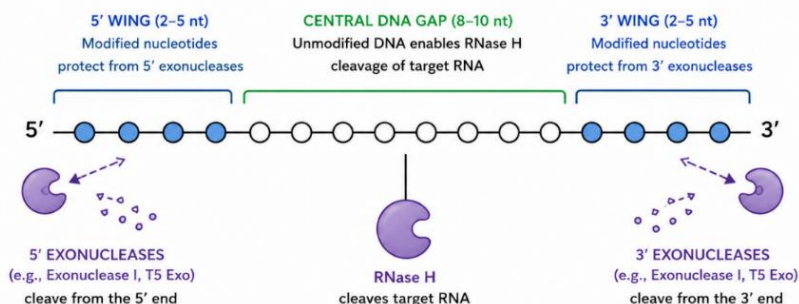
X-RAY DIFFRACTION Hybrid-binding domain of human RNase H1 in complex with 12-mer RNA/DNA (PDB 3BSU) (Nowotny, 2008)

Antisense oligonucleotide (ASO) therapeutics exploit these natural biological mechanisms for therapeutic purposes. When an ASO hybridizes to a complementary RNA transcript, it forms an RNA–DNA duplex that serves as a substrate for RNase H. The enzyme subsequently cleaves the phosphodiester backbone of the RNA strand, generating RNA fragments and preventing translation of downstream protein products.

To apply this mechanism therapeutically, scientists needed to overcome the challenge of delivering DNA-like molecules to their RNA targets while protecting them from rapid nuclease degradation and preserving their ability to recruit RNase H. This led to the development of chemical modifications that maintain the DNA-like properties required for

RNase H activity while substantially improving oligonucleotide stability in biological systems. Early unmodified oligonucleotides were highly susceptible to enzymatic degradation, severely limiting their therapeutic potential. This limitation was first addressed through backbone modifications, most notably phosphorothioate linkages, which confer resistance to nuclease-mediated degradation while retaining compatibility with RNase H activity. In parallel, sugar modifications such as 2'-O-methyl and 2'-O-methoxyethyl groups were introduced to enhance binding affinity, increase metabolic stability, and improve overall pharmacological properties.

These are often incorporated into “gapmer” designs, where a central DNA region capable of recruiting RNase H is flanked by modified nucleotides for protection. Collectively, these innovations enable antisense oligonucleotides to form stable RNA–DNA hybrids, activate RNase H, and achieve sufficient stability, bioavailability, and specificity for effective therapeutic application.



These advances in antisense chemistry and RNase H-mediated mechanisms have transformed ASOs from experimental molecular tools into precision therapeutics capable of targeting disease-causing transcripts with remarkable specificity. What began with Kossel’s early characterization of nucleic substances and the later discovery of nucleases has culminated in a new era of individualized genetic medicine, where pathogenic mutations can be selectively silenced at the RNA level while preserving normal cellular function.

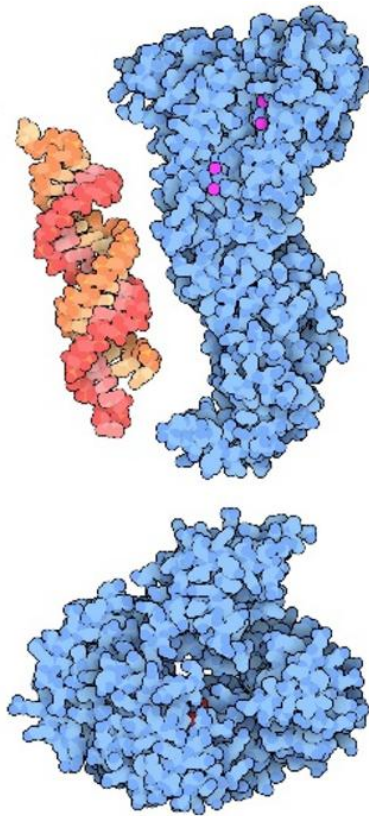
La Jolla Labs, a neurogenetics therapeutics company recently identified and developed an allele-selective antisense oligonucleotide targeting a familial form of amyotrophic lateral sclerosis (ALS). By leveraging advances in ASO chemistry the program moved from mutation identification to clinical therapeutic development in less than six months, an unprecedented timeline in the field of precision neurogenetics. The ASO

was specifically engineered to selectively suppress the mutant allele while sparing expression of the wild-type transcript, thereby minimizing the risk of disrupting essential physiological protein function.



The rapid development of this therapy illustrates how decades of fundamental discoveries in nuclease biology, RNA–DNA hybrid metabolism, and oligonucleotide engineering are now enabling highly personalized interventions for previously untreatable neurodegenerative diseases. The patient is currently being dosed at the University of Heidelberg, bringing this scientific trajectory full circle: from Kossel’s foundational investigations of nucleic acids in Heidelberg more than a century ago to the modern application of RNase H–dependent antisense therapeutics in the treatment of human disease at the same institution. This convergence of historical discovery, molecular biology, and translational medicine highlights the extraordinary progress of nucleic acid therapeutics and underscores the growing potential of allele-selective ASOs as rapid, adaptable treatments for rare genetic disorders.

RISC-Mediated Gene Silencing: RNA Interference (siRNA)



DAVID S. GOODSSELL

Small interfering RNA (siRNA) is a powerful cellular defense system that helps protect organisms from viral infections and regulate gene expression. Cells are particularly sensitive to double-stranded RNA, which is often produced during viral replication and signals potential danger. In response, specialized proteins such as dicer (see top blue protein in David Goodsell's watercolor illustration) cut these double-stranded RNA into short fragments. These are then incorporated into argonaut proteins (bottom protein in Goodsell's illustration; PDB 248S), which use one strand as a guide to identify and destroy matching messenger RNA (mRNA) molecules. Through this process, known as RNA interference, cells can selectively silence specific genes or eliminate viral RNA (Fire, A., Xu, S., Montgomery, M.K., Kostas, S.A., Driver, S.E. and Mello, C.C. 1998).

Andrew Z. Fire and Craig C. Mello received the 2006 Nobel Prize in Physiology or Medicine for this discovery of RNA interference (RNAi). Their landmark experiments, published in *Nature* in 1998, fundamentally

changed the understanding of gene regulation in eukaryotic organisms and opened entirely new avenues in molecular biology, biotechnology, and medicine.

Fire and Mello's work was conducted using the nematode *Caenorhabditis elegans* (*C. elegans*), a small transparent roundworm widely used as a model organism in genetics. At the time researchers knew that introducing "antisense" RNA could suppress gene expression. However, the effects were weak and inconsistent, and the mechanism remained unclear. Fire and Mello hypothesized that the true silencing trigger might not be single-stranded antisense RNA itself, but rather a "contaminating" double-stranded RNA formed during their experimental preparation.

To test this idea, they designed a series of carefully controlled experiments targeting the *unc-22* gene in *C. elegans*. This gene is responsible for normal muscle function; when it is disrupted, worms exhibit a characteristic twitching movement that can easily be observed under a microscope. The researchers synthesized three types of RNA corresponding to the *unc-22* sequence:

Sense RNA (matching the mRNA sequence)

Antisense RNA (complementary to the mRNA)

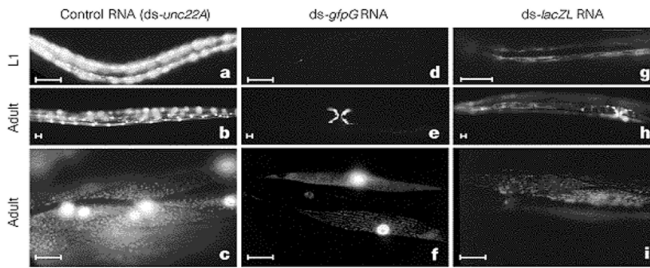
Double-stranded RNA created by hybridizing the sense and antisense strands

The results were striking; injection of either sense RNA or antisense RNA alone produced little or inconsistent gene silencing. In contrast, injections of tiny amounts of double-stranded RNA caused potent and highly specific suppression of the *unc-22* gene. The treated worms displayed the expected twitching phenotype, indicating that expression of the target gene had been effectively shut down. Importantly, Fire and Mello demonstrated that the effect was:

Sequence-specific: only genes matching the dsRNA sequence were silenced.

Highly potent: extremely small amounts of dsRNA were sufficient.

Systemic: the silencing effect spread throughout the organism.



The figure demonstrates that double-stranded RNA (dsRNA) can specifically silence gene expression in animal offspring through RNA interference. Control animals injected with ds-unc22A showed normal GFP expression in both nuclei and mitochondria, while animals injected with ds-gfpG displayed a dramatic loss of GFP expression, with only a few cells retaining activity. In contrast, ds-lacZL selectively eliminated nuclear GFP–LacZ expression without affecting mitochondrial GFP, showing that RNA interference can precisely target specific genes while leaving others unchanged.

Fire, A., Xu, S., Montgomery, M. *et al.* Potent and specific genetic interference by double-stranded RNA in *Caenorhabditis elegans*. *Nature* **391**, 806–811 (1998).
<https://doi.org/10.1038/35888>

Heritable: gene silencing persisted in subsequent generations of worms.

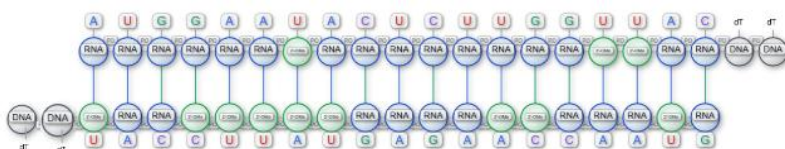
They repeated similar experiments with other genes and obtained comparable results, confirming that RNAi represented a general biological mechanism rather than a gene-specific anomaly. Their findings suggested that cells possess machinery capable of recognizing dsRNA and use it as a signal to destroy complementary mRNA molecules. Subsequent research revealed the molecular basis of this pathway. Double-stranded RNA is processed by an enzyme called Dicer into short interfering RNAs (siRNAs), typically around 21–23 nucleotides long. These siRNAs are incorporated into the RNA-induced silencing complex (RISC), which uses the siRNA sequence as a “guide” to identify and cleave matching mRNA targets, thereby preventing protein production.

RNAi has since become an important therapeutic strategy; one that enables precise silencing of disease-associated genes, including targets often considered “undruggable” by conventional therapies (Hu et al., 2020). Since the first FDA approval of Patisiran in 2018 for hereditary transthyretin-mediated amyloidosis, the field has advanced into a clinically validated therapeutic modality. Subsequent approvals of Givosiran, Lumasiran, Inclisiran, Vutrisiran, and Nedosiran expanded its reach into metabolic, cardiovascular, and rare genetic diseases. As of 2025, more than 260 siRNA drug candidates were in preclinical or clinical development. Research now spans major therapeutic areas, including oncology, neurodegenerative disease, infectious disease, and diabetes.

Despite its therapeutic promise, RNA interference (RNAi) therapy continues to face significant challenges related to the efficient delivery of

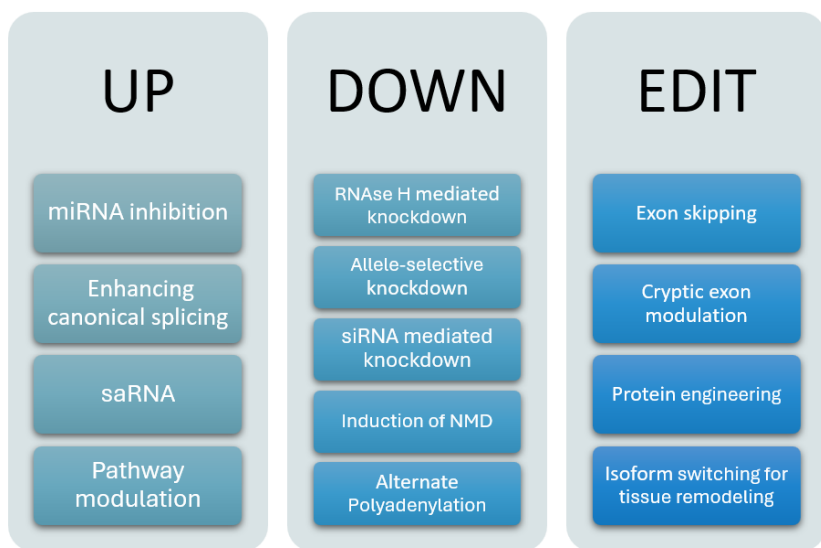
siRNAs to their intended targets. Unmodified siRNAs are highly unstable in vivo and are rapidly degraded by endogenous ribonucleases, resulting in extremely short circulation half-lives. In addition, their negatively charged, hydrophilic nature severely limits passive diffusion across cellular membranes, making intracellular delivery highly inefficient. Even when cellular uptake occurs, most siRNAs become trapped within endosomal compartments, preventing access to the cytoplasm where gene silencing takes place. Consequently, endosomal escape is widely regarded as the rate-limiting barrier restricting broader therapeutic applications of RNAi. Although numerous delivery strategies have been developed, including lipid conjugates, cationic peptides, phosphorothioate modifications, and viral delivery systems, many of these approaches are associated with toxicity caused by nonspecific endosomal disruption (Dowdy, 2023).

Onpatro (Patisiran): FDA-approved siRNA therapeutic indicated for the treatment of hereditary transthyretin-mediated amyloidosis (hATTR amyloidosis)



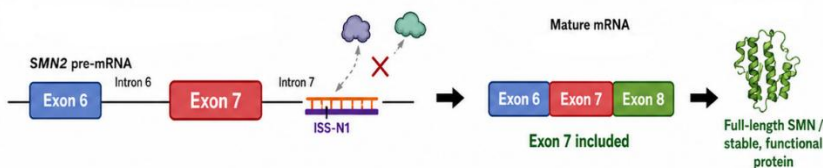
ONPATTRO (patisiran) was the first FDA-approved siRNA therapeutic indicated for the treatment of hereditary transthyretin-mediated amyloidosis (hATTR amyloidosis) with polyneuropathy. It functions through RNA interference by selectively degrading transthyretin (TTR) mRNA in hepatocytes, thereby reducing production of the pathogenic TTR protein and slowing progression of neuropathic disease.

In addition to delivery challenges, naked siRNAs can activate innate immune sensors, triggering inflammatory responses and unintended off-target gene silencing. To overcome these limitations, researchers have developed a range of sophisticated chemical modifications designed to improve siRNA stability, specificity, and pharmacokinetic behavior. Common modifications include 2'-O-methyl, 2'-fluoro, and phosphorothioate substitutions, while more advanced structural designs such as glycol nucleic acids (GNAs) further enhance binding affinity and target selectivity. Terminal conjugations, including cholesterol, GalNAc ligands, antibodies, and aptamers, have also enabled tissue-specific targeting and improved biodistribution. These advances in chemical engineering and delivery technology are transforming siRNA therapeutics into an increasingly versatile and clinically mature platform.



Steric-Blocking and Splice-Switching Oligonucleotides

Steric-blocking antisense oligonucleotides (ASOs) sit within a broader RNA-targeting therapeutic landscape in which short, programmable oligonucleotides are used to increase, decrease, or remodel gene expression. Unlike gapmer ASOs, which recruit RNase H to cleave target RNA, steric-blocking ASOs are typically fully chemically modified and act by physical occupancy of an RNA sequence, thereby preventing access by RNA-binding proteins, spliceosome components, ribosomes, microRNAs, or 3'-end processing factors (Bennett and Swayze, 2010; Havens and Hastings, 2016). This seemingly simple mode of action has proven unusually versatile: depending on where the ASO binds, it can enhance productive gene output, suppress pathogenic RNA or protein production, or edit the architecture of the mature transcript.



One major opportunity is gene or isoform UP-regulation, where ASOs are used to increase the production or activity of a beneficial protein. For example, nusinersen binds to the ISS-N1 element downstream of SMN2 exon 7 and blocks inhibitory splicing factors, thereby promoting exon 7 inclusion and increasing full-length SMN protein in spinal muscular atrophy (Hua et al., 2008; Finkel et al., 2017).

Related concepts include enhancing canonical splicing by masking cryptic splice sites or silencer elements, as well as inhibiting microRNAs through antagomirs or steric-blocking oligonucleotides that prevent miRNA–mRNA interactions (Krützfeldt et al., 2005). ASOs have also been explored as translational activators by targeting inhibitory elements in 5' UTRs, including upstream open reading frames (uORFs), with the goal of relieving translational repression and increasing downstream protein production (Liang et al., 2016). However, uORF-directed activation remains an emerging and context-dependent strategy, with recent replication studies highlighting that not all reported uORF-targeting effects are readily generalizable (Roberts et al., 2024). More broadly, these “UP” approaches suggest that ASOs can be used not only to silence genes, but also to restore insufficient protein expression or rebalance disease-associated pathways.

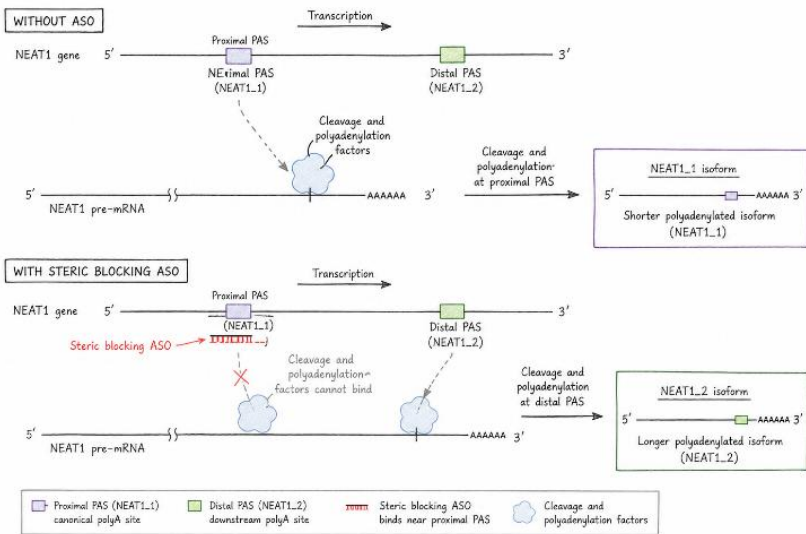
Steric blocking ASO's can also be used for gene DOWN-regulation, where oligonucleotides reduce harmful RNA or protein output. Classical antisense gapmers achieve this through RNase H–mediated cleavage of RNA–DNA hybrids, however, Steric-blocking ASOs can also reduce gene output indirectly by redirecting splicing to introduce premature termination codons and trigger nonsense-mediated decay, or by modulating alternative polyadenylation to alter transcript stability, localization, coding potential, or 3' UTR regulatory architecture (Havens and Hastings, 2016; Tian and Manley, 2017).

A third and particularly creative mode is “editing”, in which ASOs reshape the transcript rather than simply increasing or decreasing its abundance. Exon skipping in Duchenne muscular dystrophy is the most well-known clinical example: phosphorodiamidate morpholino oligomers such as eteplirsen are designed to mask splice elements in the DMD pre-mRNA, skip selected exons, restore the reading frame, and permit production of an internally deleted but partially functional dystrophin protein (Mendell et al., 2016; Aartsma-Rus and Corey, 2020). Similar splice-switching strategies can suppress cryptic exons, as shown by milasen, a personalized ASO designed to correct a patient-specific splicing defect in CLN7 disease (Kim et al., 2019). Beyond correction of single pathogenic events, ASOs can be used for protein engineering at the RNA level by controlling exon inclusion, exclusion, or isoform balance, potentially enabling tissue-specific remodeling of protein domains, signaling outputs, or developmental programs (Havens and Hastings, 2016; Lim et al., 2020).

One particularly creative transcript-editing strategy uses steric-blocking ASOs to remodel isoform output of NEAT1 with the goal to shift expression away from the shorter NEAT1_1 isoform and toward the longer

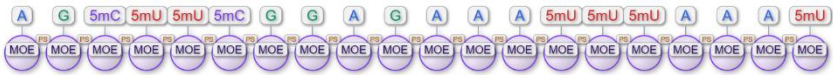
NEAT1_2 isoform. The two isoforms have different biological functions: NEAT1_2 acts as an architectural long non-coding RNA that scaffolds paraspeckles, nuclear bodies that can sequester RNA-binding proteins and regulate gene-expression programs. By contrast, NEAT1_1 does not efficiently form paraspeckles and can support paraspeckle-independent functions that, in some cancer contexts, are associated with increased cell proliferation. In high-risk neuroblastoma models, this isoform balance appears functionally important where NEAT1_1 is linked to a more proliferative cancer-cell state; increased NEAT1_2 and paraspeckle formation are associated with reduced neuroblastoma cell viability and activation of differentiation-related pathways.

In 2021, Naveed and colleagues at the University of Western Australia described a transcript-editing strategy that used steric-blocking ASOs to manipulate alternative polyadenylation of the long non-coding RNA NEAT1. Working in neuroblastoma models, the researchers designed ASOs that bind sequences surrounding the canonical polyadenylation site of the short NEAT1_1 isoform. By occupying this region, the ASOs physically blocked cleavage and polyadenylation factors from processing the transcript at the proximal polyadenylation site. This forced transcription to continue beyond the usual NEAT1_1 termination site and promoted production of the longer NEAT1_2 isoform through a downstream processing pathway. The study showed that steric-blocking ASOs can redirect RNA 3'-end formation, rather than simply alter splicing, and demonstrated how isoform switching can be used to probe opposing biological functions of long non-coding RNA isoforms in cancer cells (Naveed et al., 2021).



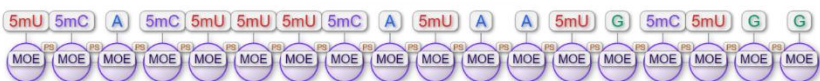
Naveed, A., Cooper, J.A., Li, R., Hubbard, A., Chen, J., Liu, T., Wilton, S.D., Fletcher, S. and Fox, A.H. (2021) 'NEAT1 polyA-modulating antisense oligonucleotides reveal opposing functions for both long non-coding RNA isoforms in neuroblastoma', *Cellular and Molecular Life Sciences*, 78, pp. 2213–2230. Available at: <https://doi.org/10.1007/s00018-020-03632-6>

Splice switching ASO for Usher syndrome type 2



Ulteversen (QR-421a) is a 21-mer chemically modified splice-altering ASO for the treatment of USH2A-associated retinitis pigmentosa and Usher syndrome type 2. The molecule incorporates a fully phosphorothioate backbone together with all nucleosides modified with 2'-O-(2-methoxyethyl; aka MOE) sugars (Girach et al., 2022). Ulteversen binds selectively to USH2A pre-mRNA and induces exon 13 skipping, thereby restoring production of a shorter but functional usherin protein (Girach et al., 2022). This mechanism is particularly relevant for USH2A disease because the large size of the USH2A gene limits conventional AAV-mediated gene replacement strategies. Intravitreal administration allows broad retinal distribution without the need for subretinal surgery or viral vectors; reducing procedural and inflammatory risks associated with gene therapy (Girach et al., 2022). Preclinical studies demonstrated sustained exon skipping, restoration of usherin expression, retinal uptake across photoreceptor layers, and recovery of functional electroretinographic responses (Dulla et al., 2021). Clinically ulteversen demonstrated a favorable tolerability profile with no serious ocular adverse events or treatment discontinuations, alongside trends toward stabilization of retinal structure and visual function compared with sham treatment (Yang et al., 2025). Preservation of ellipsoid zone area, together with stabilization of visual acuity and retinal sensitivity measures, suggests that ulteversen may slow progressive photoreceptor degeneration in USH2A-associated RP, supporting continued evaluation in ongoing Phase 2b/3 studies (Yang et al., 2025).

Splice modulating ASO to generate a productive protein product

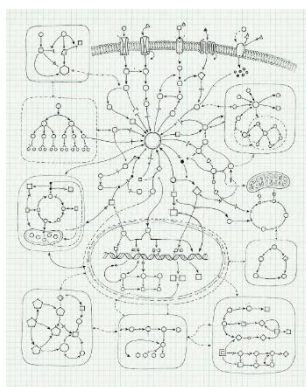


Another splicing example, Spinraza (nusinersen) is also a fully modified antisense oligonucleotide with a phosphorothioate backbone and fully modified 2'-O-methoxyethyl sugar 18mer. This ASO enhances the stability of the SMN2 pseudogene and is clinically indicated for the treatment of spinal muscular atrophy (SMA) in both pediatric and adult patients. The dosing regimen consists of four loading doses (three given at 14-day intervals followed by a fourth after 30 days), followed by a 12 mg

maintenance dose every four months (Biogen, 2016) illustrating the therapeutic robustness of this compound in the CNS.

Beyond the approved therapies for SMA and Duchenne muscular dystrophy, many splice-switching antisense oligonucleotides (ASOs) are now in preclinical and clinical development across a wide range of diseases. Active programs target rare neuromuscular and neurodevelopmental disorders, including Dravet syndrome, Batten disease, ataxia-telangiectasia, Huntington's disease, as well as emerging applications in cancer and immune-mediated disorders.

Splice-Switching for Pathway Modulation



Splice-switching ASOs can also be used as pathway-modulating therapeutics, where the objective is to reshape the behavior of a disease-relevant signaling network rather than simply increase or decrease expression of a single gene. This strategy is especially powerful when systems-level analysis identifies a regulatory node whose activity depends on isoform structure, protein–protein interactions, feedback regulation, or subcellular recruitment. In these cases, the most therapeutically useful intervention may not be gene silencing, but isoform rebalancing: shifting RNA splicing toward a version of the protein that preserves essential function while reducing pathogenic pathway output. Such approaches align naturally with systems biology, because they treat disease pathways as dynamic circuits with tunable control points. Instead of removing a component from the network, splice-switching can alter how that component transmits signal through the pathway. The SOS1/RAS axis provides a useful example of this logic, because SOS1 sits upstream of RAS activation and its splice isoforms appear to differ in Ras-stimulatory activity. A splice-switching strategy that favors the less activating SOS1 isoform could therefore reduce RAS–MAPK signaling while maintaining overall

SOS1 expression and avoiding the broader consequences of complete SOS1 knockdown.

Splice-switching to attenuate RAS activity through SOS1 isoform switch

RAS signaling is one of the central pathways controlling cell growth, survival, differentiation, and proliferation. In cancer, this pathway is frequently hyperactivated, either through activating mutations in RAS itself or through upstream regulators that increase the amount of active, GTP-bound RAS. One such upstream regulator is SOS1, a guanine nucleotide exchange factor that promotes conversion of inactive RAS-GDP into active RAS-GTP. Because SOS1 sits upstream of RAS activation, changing SOS1 function offers a potential way to reduce signaling through the RAS–MAPK pathway.

A group of scientists at La Jolla Labs designed a splice-switching strategy around a naturally occurring alternative splicing event in SOS1. SOS1 encodes a widely expressed guanine nucleotide exchange factor (GEF) that activates RAS-family GTPases by promoting GDP–GTP exchange, thereby increasing the pool of active RAS-GTP and downstream signaling through pathways such as RAS–RAF–MEK–ERK/MAPK (Baltanás et al., 2020). Human cells express two major SOS1 splice isoforms that differ by inclusion or exclusion of a short 45 bp exon, referred to here as exon 21. The exon-included isoform is referred to as IsfII, while the exon-skipped isoform is referred to as IsfI. Prior work identified these two human SOS1 isoforms and reported that they differ by a 15-amino-acid segment in the C-terminal proline-rich region, a regulatory region involved in adaptor-protein binding and control of SOS1 activity (Baltanás et al., 2020). That work suggested that inclusion of this segment enhances the Ras-stimulatory activity of SOS1. In this model, the +exon 21 isoform, IsfII, is more active, whereas the –exon 21 isoform, IsfI, is predicted to be less activating toward RAS.

The therapeutic idea, therefore, is not to knock down SOS1 entirely, but to bias splicing away from the more RAS-activating isoform and toward the less activating isoform. Splice-switching oligos were designed to sterically block inclusion of exon 21, forcing the transcript to skip this exon. By reducing production of IsfII and increasing production of IsfI, the oligos are expected to attenuate SOS1-mediated RAS activation while preserving overall SOS1 expression.

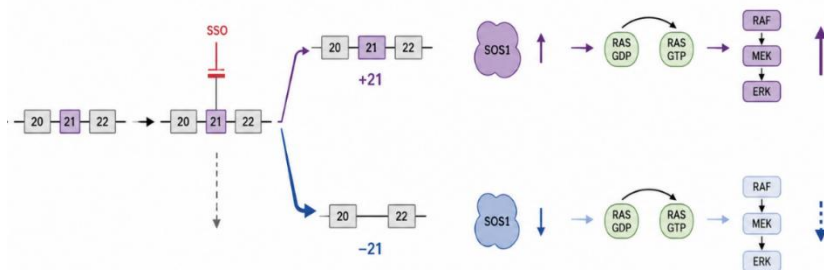
This distinction is important because SOS1 is not simply an abundance-driven target. Its activity is controlled by domain architecture, membrane recruitment, adaptor binding, autoinhibition, and allosteric

activation. SOS1 contains N-terminal regulatory domains, including histone-like, Dbl homology and pleckstrin homology domains, a central catalytic module containing REM and CDC25H domains, and a C-terminal proline-rich region that mediates interactions with SH3-domain-containing adaptor proteins such as GRB2 and E3B1 (Baltanás et al., 2020). Total SOS1 abundance may therefore remain largely unchanged while the functional composition of SOS1 isoforms changes; i.e. to alter the ratio of SOS1 splice isoforms in favor of a form with lower Ras-stimulatory activity. This makes the strategy conceptually different from siRNA, CRISPR knockout, or degradation-based approaches. It is a functional isoform switch, not a simple loss-of-expression strategy.

Mechanistically, SOS1 promotes RAS signaling by acting as a GEF. It helps exchange GDP for GTP on RAS, thereby increasing active RAS-GTP. Active RAS then propagates signaling through downstream pathways, including the RAS–MAPK cascade. SOS1 activity is also shaped by membrane localization, GRB2-mediated recruitment to activated receptor tyrosine kinases, allosteric stimulation by RAS-GTP, and negative feedback through ERK- and RSK-mediated phosphorylation of the SOS1 C-terminal region (Baltanás et al., 2020). In resting cells, SOS1 is maintained in an autoinhibited state through regulatory contributions from both the N-terminal and C-terminal regions. Recruitment to the membrane and interaction with adaptor proteins can relieve this inhibition, allowing SOS1 to engage membrane-localized RAS and catalyze nucleotide exchange.

The C-terminal proline-rich region is especially relevant to the splice-switching strategy because it participates in adaptor binding and regulatory control. GRB2–SOS1 complexes recruit SOS1 to activated receptor tyrosine kinases, enabling RAS activation at the inner plasma membrane. Alternative splicing within this region could therefore alter SOS1 signaling output since it targets a regulatory region rather than the catalytic core itself.

The splice-switching strategy: reduce RAS pathway activation by shifting SOS1 splicing from the exon 21-included isoform to the exon 21-skipped isoform and decreased generation of active RAS-GTP with further reduced signal flow through the RAS–MAPK pathway, while avoiding complete loss of SOS1 (Baltanás et al., 2020).



Experimental results support the feasibility of this splicing approach. The lead oligos showed upregulation of the *IsfI* isoform and loss or strong reduction of detectable *IsfII*. Importantly, general *SOS1* abundance was largely maintained, except for some exon 22-targeting designs that increased total *SOS1* signal. Sanger sequencing was consistent with the qPCR results, supporting the interpretation that the observed changes reflected genuine splice switching rather than assay artefact.

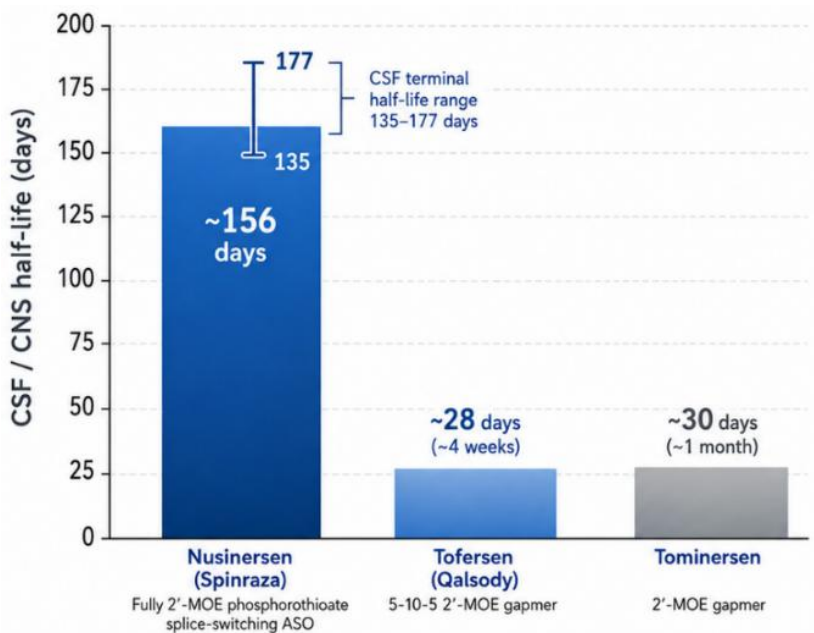
The main unresolved biological question is whether forcing *SOS1* toward *IsfI* is sufficient to produce a meaningful reduction in RAS signaling in disease-relevant cells. Key next steps include dose-response testing of lead oligos, validation in additional cancer cell lines, and protein-level readouts of RAS pathway activity. Useful assays could include measurements of RAS-GTP, phospho-MEK, phospho-ERK, or other downstream MAPK signaling markers. It will also be important to determine whether this strategy works similarly across *KRAS*-, *HRAS*-, and *NRAS*-driven contexts, and whether it remains effective in cells carrying activating RAS mutations. Because *SOS1* can also influence *RAC1*-dependent processes through adaptor-specific signaling complexes, assays of migration, cytoskeletal phenotypes, or *RAC1* activation may also be useful in selected contexts.



Primary screen of SOS1 isoform switch ASOs Group 2 ASOs show upregulation of ISF1 & no detection of ISF2
Overall SOS1 general abundance stays the same (except Exon 22 ASOs – Group 2 – is upregulated)
Sanger sequencing correlates to qPCR results

Overall, these splice-switching oligos were designed as a targeted way to attenuate RAS activity by exploiting a functional SOS1 splicing event. By blocking exon 21 inclusion, the oligos promote production of a less RAS-stimulatory SOS1 isoform while largely preserving total SOS1

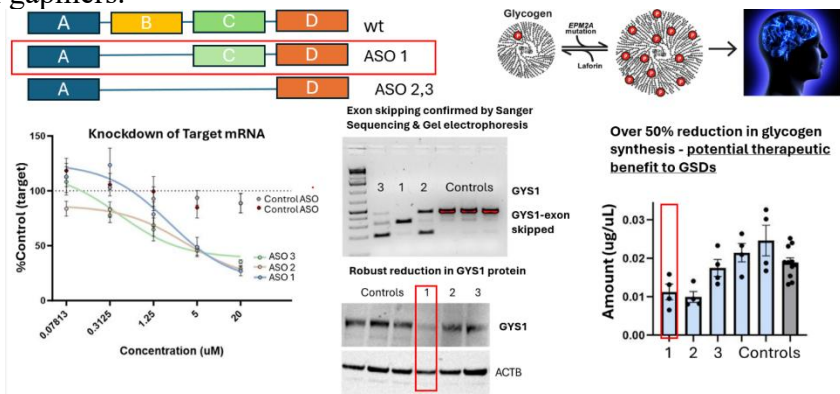
expression. This makes SOS1 exon 21 skipping an attractive model for splice-switching therapeutic development: the biology is tied to a clear functional isoform change, the mechanism acts upstream of RAS activation, and the pharmacologic intervention is well matched to the splicing-based mode of action. More broadly, the strategy fits with the emerging view of SOS1 as both a regulator of normal RAS signalling and a therapeutic target in RAS-dependent disease, but it seeks to modulate SOS1 function through isoform switching rather than direct enzymatic inhibition or protein depletion.



Gene Knockdown Using Nonsense Mediated Decay

RNase H-mediated knockdown via gapmer technologies such as tofersen use a chimeric architecture, typically with modified 2'-MOE “wings” flanking a central DNA gap, which enables RNase H recruitment and degradation of the target RNA. This design is potent, but the central DNA gap is less nuclease-protected than a fully modified steric-blocking ASO. Tofersen and Tominersen are 20-mer 5-10-5 MOE gapmers, with an estimated effective CSF half-life of approximately four weeks when intrathecally delivered (FDA, 2023; Goyal et al., 2023). By contrast, Nusinersen, a fully modified 2'-MOE phosphorothioate splice-switching ASO that does not require a DNA gap or RNase H recruitment has a substantially greater nuclease resistance and a much longer CSF terminal half-life of approximately 135–177 days (FDA, 2026).

This distinction suggests an attractive therapeutic strategy for targets where knockdown can be achieved by splice modulation rather than RNase H cleavage: a fully modified ASO that redirects splicing to create a premature termination codon and thereby induces nonsense-mediated decay (NMD). The drug behaves pharmacokinetically like nusinersen, but pharmacodynamically it reduces target protein by forcing the transcript into a non-productive, NMD-sensitive isoform. Therefore, when the biology of the target permits NMD-mediated silencing, a fully modified splice-switching ASO could offer a superior profile for chronic neurological disease: durable CNS exposure, less frequent intrathecal dosing, and avoidance of the shorter half-life liability inherent to DNA-gap-containing RNase H gapmers.



Fully modified steric-blocking ASO chemistry offers a compelling advantage when the therapeutic mechanism does not require RNase H recruitment. By eliminating the central DNA gap, these molecules are inherently more resistant to nuclease degradation, supporting longer residence time in the CNS and potentially enabling less frequent intrathecal dosing. The contrast between gapmer ASOs such as tofersen and tominersen, which rely on a nuclease-vulnerable DNA gap, and fully modified splice-switching ASOs such as nusinersen highlights the pharmacological value of chemical stability in chronic neurological indications (FDA, 2023; Goyal et al., 2023; FDA, 2026). For targets amenable to splice redirection and NMD-mediated silencing, this approach could combine the durability of fully modified 2'-MOE phosphorothioate chemistry with the functional outcome of protein knockdown. Thus, fully modified ASOs may provide a particularly attractive therapeutic architecture: nuclease-resistant, long-lasting, tunable, and mechanistically capable of reducing pathogenic protein expression without the shorter half-

life liability and tighter therapeutic window associated with DNA-gap-containing RNase H gapmers.

Optimize Productive Splicing for Gene Upregulation

Many genetic diseases arise not from toxic gain-of-function mutations, but from insufficient production of a normal or functional protein. These haploinsufficient disorders occur when one working copy of a gene does not produce enough protein to maintain normal cellular function. Traditional therapeutic strategies such as gene replacement, genome editing, recombinant protein delivery, or modified RNA can be powerful, but they may be limited by gene size, tissue targeting, immune responses, durability, or difficulty controlling expression levels. These challenges have created interest in approaches that can increase expression of a patient's own endogenous gene in a selective and titratable manner.

Antisense oligonucleotides, or ASOs, provide one such strategy. ASOs are short, chemically modified nucleic acids designed to bind specific RNA sequences through base pairing. Depending on their chemistry and design, they can reduce RNA levels, alter RNA processing, or redirect pre-mRNA splicing. While many ASO therapies have historically focused on silencing toxic transcripts or correcting mutation-specific splicing defects, a newer concept uses ASOs to increase gene expression by preventing naturally occurring non-productive splicing events.

This approach was demonstrated by Stoke Therapeutics in 2020 (Lim et al., 2020). The company introduced a platform called TANGO, or targeted augmentation of nuclear gene output, which uses ASOs to block alternative splicing events that normally generate premature-termination-codon-containing transcripts targeted for nonsense-mediated mRNA decay. By preventing these non-productive splice forms, the cell produces more productive mRNA, leading to increased full-length protein from the native gene locus.

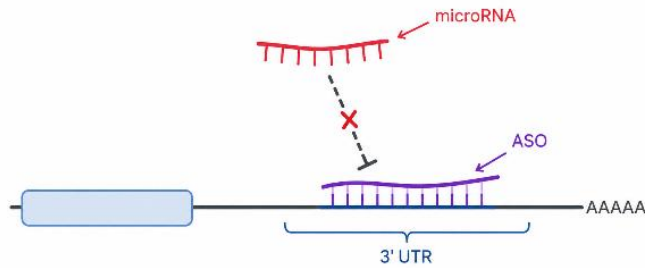
In the study, Stoke Therapeutics analyzed human RNA-seq datasets from ASO-accessible tissues, including liver, kidney, central nervous system, and eye, and identified non-productive splicing events in 7,757 protein-coding genes. Of these, 1,246 were associated with genetic disease. The authors then tested the strategy across multiple genes and splicing-event classes, including PCCA, SYNGAP1, CD274, and SCN1A. These genes represented different biological functions, disease contexts, and coding-sequence sizes, ranging from CD274 at 870 bp to SCN1A at 6,015 bp.

The work showed that ASOs could dose-dependently reduce non-productive splicing and increase productive mRNA in vitro. Importantly, this mRNA increase translated into higher protein levels, with reported

protein upregulation of approximately 1.5- to 2.6-fold at the highest ASO concentrations. Stoke also demonstrated in vivo proof of concept by delivering SCN1A-targeting ASOs into neonatal mouse brain, where treatment increased productive Scn1a mRNA and NaV1.1 protein in a dose-dependent manner.

Together, these findings established ASO-mediated upregulation as a mutation-independent strategy for increasing endogenous protein expression. Rather than replacing a gene or editing DNA, this approach leverages natural RNA-processing biology to shift splicing away from non-productive transcripts and toward productive mRNA.

Blocking miRNA Binding Sites for Gene Upregulation

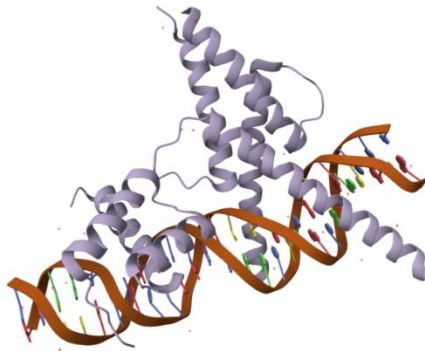


MicroRNAs are small endogenous non-coding RNAs that usually repress gene expression by binding target transcripts and reducing mRNA stability or translation (Bartel, 2018; Shang et al., 2023). MicroRNAs often have multiple targets, can act differently depending on cell state, and may produce indirect effects through regulatory networks. microRNA-based upregulation strategies require careful validation at multiple stages before their effects can be interpreted therapeutically. First, the candidate microRNA must be shown to be expressed in the disease-relevant tissue or cell type, since tissue-specific microRNA expression and tissue-restricted target availability strongly influence whether a given microRNA–mRNA interaction is biologically active (Sood et al., 2006; Liang et al., 2009; Liang and Li, 2011). This is particularly important because a microRNA that produces an apparent regulatory effect in one tissue may have little or no activity in another if the microRNA itself, its cofactors, or its relevant targets are absent. Second, during in vitro screening, the selected cell lines must be confirmed to express the microRNA of interest at physiologically

meaningful levels, rather than if a general cellular model recapitulates the relevant tissue context. Even when expression is confirmed, quantifying microRNA-mediated upregulation without bias remains challenging, because observed increases in target gene expression may reflect direct activation, indirect network effects, differences in baseline transcript abundance, or cell-state-dependent shifts in microRNA function. These considerations make it difficult to assign a precise and generalizable magnitude of upregulation during the screening phase and underscore the need for orthogonal validation in tissue-relevant models

microRNA targeting for up regulation of TCF4 haploinsufficiency

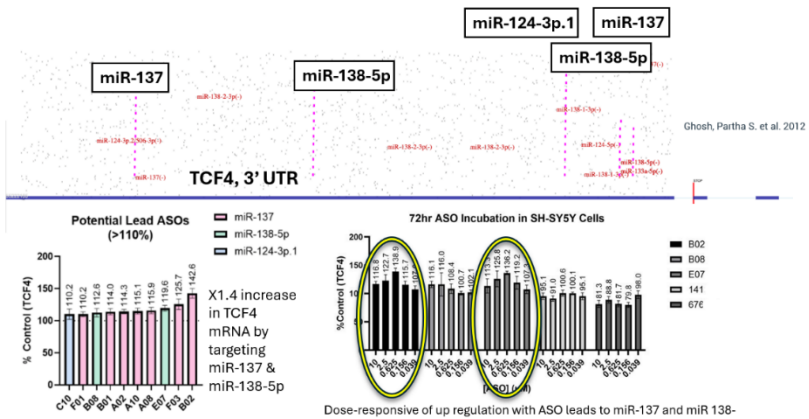
Pitt-Hopkins syndrome, or PTHS, is a rare neurodevelopmental disorder most often caused by a heterozygous pathogenic variant in TCF4 or by a deletion of chromosome 18q21.2 involving TCF4 (Zollino et al., 2019; Whalen et al., 2012). Clinically, PTHS is associated with developmental delay, moderate-to-severe intellectual disability, severely limited or absent speech, motor delay, seizures, constipation, sleep disturbance, and characteristic breathing abnormalities such as episodic hyperventilation or breath-holding



(Zollino et al., 2019; Bedeschi et al., 2017). Molecular diagnosis is established by identifying a heterozygous pathogenic or likely pathogenic TCF4 variant, small intragenic deletion, or larger 18q21.2 deletion involving the gene (Sweetser et al., 2024).

The core disease mechanism in many cases is TCF4 haploinsufficiency. This means that one functional copy of TCF4 does not produce enough *transcription factor 4* protein to support normal neurodevelopmental gene-regulatory programs. TCF4 encodes a class I

basic helix-loop- helix transcription factor (PDB: 8OSB), and disruption of its dosage affects broad downstream transcriptional networks relevant to neuronal development and function (Sepp et al., 2012; Dennis et al., 2019). Because PTHS is caused by reduced TCF4 function rather than by a toxic gain of function, a central therapeutic idea is to increase expression from the remaining functional allele. Preclinical work supports this logic: restoration of Tcf4 expression in a mouse model improved behavioral, electrophysiological and transcriptional phenotypes, suggesting that normalizing TCF4 dosage could be therapeutically meaningful (Kim et al., 2022).



This dosage-correction model makes PTHS an attractive candidate for RNA-based upregulation strategies. However, because TCF4 is a transcription factor with widespread effects, the therapeutic goal is not uncontrolled overexpression. The goal is to control a physiological range. ASOs, are well suited to this type of approach because they can be designed to bind RNA and alter gene expression through steric blocking (Bennett and Swayze, 2010; Ward et al., 2014).

La Jolla Labs, in collaboration with the Pitt Hopkins Research Foundation, screened antisense oligonucleotides to assess whether they could restore TCF4 protein levels by upregulating the functional copy of TCF4 in Pitt-Hopkins syndrome (PTHS) (Pitt Hopkins Research Foundation, 2023). La Jolla Labs' strategy was to disrupt microRNA-mediated repression as a way to increase endogenous TCF4 expression. MicroRNAs are short non-coding RNAs that bind target sites, often within the 3' untranslated regions of mRNAs, and reduce gene expression by promoting transcript destabilization or inhibiting translation (Bartel, 2009). If specific miRNAs repress TCF4, then blocking those miRNAs or their binding sites could relieve repression and increase TCF4 expression. Steric-blocking ASOs can, in principle, reduce miRNA activity either by binding the miRNA itself or by blocking a miRNA binding site on a particular target

mRNA, thereby preventing the miRNA-loaded silencing complex from engaging that transcript (Roberts et al., 2024).

The La Jolla Labs approach could therefore be framed as a target-site validation and pharmacological de-repression strategy. Candidate miRNA binding sites in the TCF4 transcript were first identified, but these candidates were not advanced based solely on computational prediction. Each candidate miRNA was validated for expression in the relevant tissue or cellular model. This expression filter was essential because a predicted miRNA–TCF4 interaction was only therapeutically useful if the miRNA was present in the disease-relevant context and computationally the 3' UTR contains over 1000 miRNA hits.

After tissue-expression validation, the next step was functional validation. Candidate miRNAs were tested for interaction with TCF4 to determine whether they measurably repressed TCF4 RNA or protein. Inhibition experiments then assessed whether blocking the miRNA or its binding site produced TCF4 upregulation. A dose-responsive upregulation was measured to distinguish specific effects from noise and provided a foundation potency and therapeutic-window optimization.

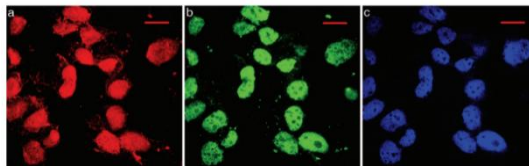
Pitt-Hopkins syndrome is a genetic disorder in which understanding the underlying disease mechanism can guide therapeutic design. Because PTHS results from TCF4 haploinsufficiency, increasing TCF4 expression represents a potential therapeutic goal. However, TCF4 is dosage-sensitive, and excessive upregulation could potentially be harmful. Unlike gene therapy, ASO-based approaches may offer a more tunable strategy for achieving an optimal level of TCF4 restoration. La Jolla Labs' miRNA-site-directed approach aimed to identify endogenous miRNA-mediated repression of TCF4 and relieve that repression using oligonucleotides. By integrating tissue-relevant miRNA expression, experimental validation of miRNA–TCF4 interaction, and dose-responsive TCF4 upregulation after inhibition, this strategy provided a rational framework for controlled restoration of TCF4 protein levels in PTHS.

MicroRNAs may also upregulate gene expression at the transcriptional level through promoter-associated mechanisms. Place et al. (2008) demonstrated that miR-373 could induce expression of CDH1, encoding E-cadherin, and CSCD2 by targeting complementary promoter sequences. This form of microRNA-mediated gene activation overlaps conceptually with RNA activation, or RNAa, in which small RNAs directed to promoter or promoter-associated sequences can increase transcription rather than silence gene expression (Place et al., 2008; Li et al., 2025). These observations expanded the view of microRNAs from purely repressive regulators to context-dependent modulators capable of either decreasing or increasing gene output.

Short Activating RNAs (saRNA)

Short activating RNAs (saRNAs) are a class of small double-stranded RNA molecules that upregulate gene expression by targeting promoter regions and triggering RNA activation (RNAa). In contrast to siRNAs, saRNAs work within the nucleus, often involving Argonaute proteins, to enhance transcription of specific genes.

A conundrum might seem that siRNAs are understood to function in the cytoplasm, whereas saRNAs, with the same size act within the nucleus to regulate transcription. Early studies of RNA activation infer nuclear localization indirectly, showing that promoter-targeted dsRNAs induce locus-specific chromatin modifications (Li et al., 2006; Janowski et al., 2007). Direct imaging studies demonstrate that siRNAs can enter the nucleus in a selective, target-dependent manner, localizing to nuclear RNAs such as 7SK (RNA polymerase II transcription elongation) and U6 (an RNA that is part of the spliceosome) (Berezhna et al., 2006). However, this is balanced by evidence that small RNAs are not freely distributed and may be actively excluded from parts of the nucleus, indicating regulated trafficking rather than constitutive nuclear access (Ohrt et al., 2006). Together, these findings suggest saRNA activity is context-dependent and maintains selective localization, rather than an absolute compartmental separation.



7SK siRNA accumulates in the nucleus in Huh-7 cells 2 h after transfection. siRNA antisense (red) and sense (green) strands colocalize primarily in the nucleus (blue) and seen in the cytoplasm only in small amounts

Berezhna, S.Y., Supekova, L., Schultz, P.G. and Deniz, A.A. (2006) 'siRNA in human cells selectively localizes to target RNA sites', *Proceedings of the National Academy of Sciences*, 103(20), pp. 7682–7687.

Because of this unique ability, saRNAs are emerging as promising tools for therapeutic gene activation (Desaulniers *et al.*, 2025). MTL-CEBPA (CEBPA-51/AW1-51), a liposomal encapsulated 21-nt saRNA targeting the transcription factor CEBPA, which is frequently silenced in hepatocellular carcinoma. In a first-in-human Phase I study (NCT02716012), MTL-CEBPA demonstrated an acceptable safety profile alongside evidence of target gene activation, antitumor activity, including

disease stabilization and occasional objective responses. These results supported advancement into randomized Phase II studies and combination trials with immunotherapies, establishing MTL-CEBPA as a prototype RNAa therapeutic and demonstrating the feasibility of systemic gene activation in humans (Reebye et al., 2018).



Another cancer-targeting RNAa therapeutic is RAG-01 (Ractigen Therapeutics), which targets the tumor suppressor CDKN1A (p21). RAG-01 is delivered via intravesical administration, a method where the drug is delivered directly into the bladder through a catheter, to patients with non-muscle invasive bladder cancer (NCT06351904). This compound attempts to directly activate p21, a historically “undruggable” pathway. Early Phase I data indicate favorable safety, absence of dose-limiting toxicities, and encouraging preliminary efficacy signals, including complete responses in carcinoma in some patients (Kang et al., 2024).

miR-551b-3p is a human microRNA which binds to a promoter and activates the expression of the STAT3 gene. miR-551b-3p activates, rather than represses, gene expression through promoter-associated RNA activation. Experimental studies in ovarian and triple-negative breast cancer models demonstrated that miR-551b-3p translocates to the nucleus and interacts with the STAT3 promoter, leading to recruitment of transcriptional machinery including RNA polymerase II and TWIST1, and increasing STAT3 transcription and downstream oncogenic signaling (Parashar et al., 2016; Parashar et al., 2019). The nuclear localization of miR-551b-3p depends on Importin-8 (IPO8), a member of the karyopherin/importin family previously identified as a nuclear transport mechanism for Argonaute-microRNA complexes (Wei et al., 2014; Parashar et al., 2019). Promoter-directed activation by RNA remains a relatively specialized and emerging area of RNA therapeutics, however the nucleotide chemistry modifications to date have not deviated from those commonly used by other platforms like siRNA.

The origins of the field can be traced to work on catalytic RNA (ribozymes), particularly by Professor Gerald Joyce and colleagues at the Salk Institute in La Jolla, California. Joyce argued that RNA possessed both “genotypic and phenotypic properties” and was therefore “a suitable substrate for molecular evolution in the laboratory” (Joyce, 1989). In his 1989 paper *Amplification, mutation and selection of catalytic RNA*, Joyce outlined a conceptual framework in which RNA molecules could undergo iterative cycles of amplification, mutation and selection, effectively creating artificial Darwinian evolution in a test tube (Joyce, 1989).

In 1990, Debra Robertson and Gerald Joyce at the Scripps Research Institute demonstrated one of the earliest laboratory evolution experiments involving RNA enzymes. Their paper, *Selection in vitro of an RNA enzyme that specifically cleaves single-stranded DNA*, showed that ribozymes could undergo repeated rounds of selection and amplification to evolve new catalytic activities (Robertson and Joyce, 1990). Importantly, they proposed that “controlled evolution experiments” could be used to generate RNAs with “novel catalytic function” (Robertson and Joyce, 1990). This work demonstrated the experimental feasibility of iterative molecular selection and helped establish the conceptual foundation for later aptamer selection technologies. Building on these advances, researchers soon began exploring whether similar evolutionary strategies could be applied not only to catalytic RNAs, but also to nucleic acids capable of highly specific molecular recognition. These efforts ultimately led to the development of in vitro selection methods capable of screening vast random-sequence nucleic acid libraries for ligand-binding activity.

In 1990 Andrew Ellington and Jack Szostak, working at the Department of Molecular Biology at Massachusetts General Hospital and Harvard Medical School in Boston, demonstrated that vast libraries of random RNA molecules could be screened to isolate sequences capable of binding small organic ligands with high specificity (Ellington and Szostak, 1990). In their landmark Nature paper, *In vitro selection of RNA molecules that bind specific ligands*, Ellington and Szostak showed that RNA populations containing approximately 10^{13} – 10^{15} distinct sequences could yield rare molecules that folded into highly specific ligand-binding structures.

At almost the same time, Craig Tuerk and Larry Gold at the University of Colorado developed the methodology that became known as SELEX (Systematic Evolution of Ligands by EXponential enrichment), formally establishing iterative selection and amplification as a general platform for evolving high-affinity nucleic acid ligands in vitro (Tuerk and

without disrupting the core folding architecture. As highlighted in broader aptamer reviews, pegaptanib exemplifies how structured oligonucleotides can achieve antibody-like specificity through well-defined three-dimensional folds, despite being identified and optimized primarily through functional selection rather than structure-first design (Keefe, Pai and Ellington, 2010).



Izervay (avacincaptad pegol) is a pegylated RNA aptamer developed for the treatment of geographic atrophy secondary to age-related macular degeneration (AMD). Unlike earlier retinal therapies that primarily targeted abnormal blood vessel growth, Izervay works by inhibiting complement component C5, a key mediator in the inflammatory complement cascade implicated in retinal cell degeneration. By selectively blocking C5 activation, the drug is designed to reduce downstream inflammatory damage and slow the progressive loss of retinal tissue associated with geographic atrophy. Its approval marked an important advance in ophthalmology, representing one of the first therapies specifically aimed at modifying the complement-driven pathology underlying dry AMD.

Aptamer Bio-tools

In addition to therapeutic aptamers, fluorogenic RNA aptamers have emerged as transformative tools for visualizing RNA localization and dynamics in living cells, enabling direct observation of RNA trafficking, translation, and phase separation with high spatiotemporal resolution. Early studies of mRNA localization using in situ hybridization and live-cell tagging systems established that asymmetric RNA distribution is a conserved mechanism for spatial regulation of gene expression across organisms ranging from yeast to neurons. Aptamer technology was quickly recognized as a tool for visualizing these dynamic processes.

The development of genetically encoded RNA imaging systems, particularly the MS2 bacteriophage platform, enabled the first “live” visualization of endogenous RNA transport dynamics in yeast and mammalian cells, revealing that localized mRNAs undergo directed, motor-dependent transport, anchoring, and translational repression during transit. Subsequent fluorogenic aptamer technology called Spinach and Mango introduced “conditional” fluorescence activation, greatly expanding utility to specific cellular contexts (Paige et al., 2011; Dolgosheina et al., 2014).

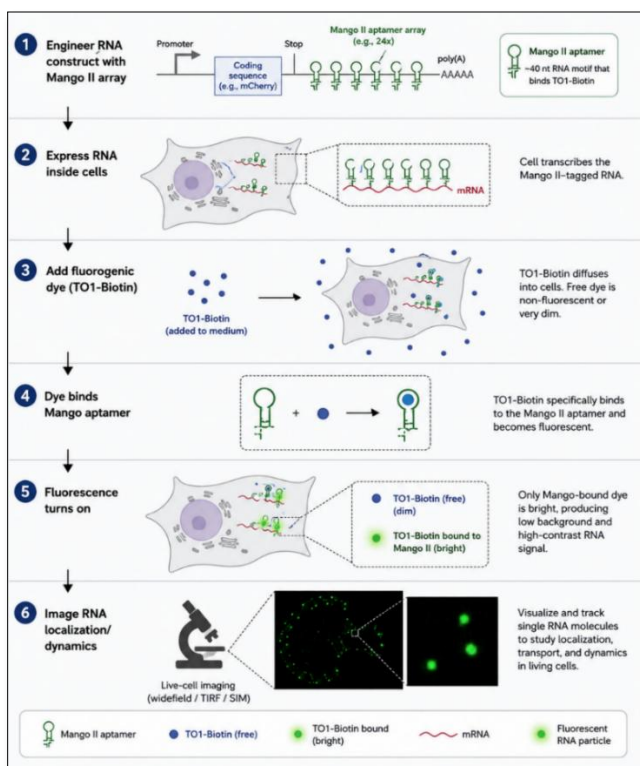
Fluorophore aptamers have also been instrumental in uncovering principles of RNA-mediated phase separation and stress granule biology. Using live-cell imaging approaches, these tools enabled visualization of mRNAs dynamically shuttling into and out of stress granules and demonstrated that RNAs themselves actively contribute to biomolecular condensate assembly rather than passively accumulating within them (Khong et al., 2017). Such studies established that selective RNA recruitment, transport kinetics, and localized translation are tightly coupled to condensate dynamics and cellular stress responses. Importantly, aberrant stress granule assembly and defective RNA trafficking have been directly linked to neurodegenerative diseases including amyotrophic lateral sclerosis (ALS) and frontotemporal dementia through pathological aggregation of RNA-binding proteins such as TDP-43 and FUS, which disrupt normal bidirectional RNA transport and granule dynamics (Williams et al., 2016; Khong et al., 2017).

Like Spinach, Mango II functions through a structurally preorganized RNA fluorophore pocket built around a compact G-quadruplex scaffold, but it achieves fluorescence activation through binding of thiazole orange derivatives such as TO1-biotin with exceptionally high affinity (~1 nM). Structural studies revealed that Mango II forms a three-tiered G-quadruplex in which the fluorophore stacks directly on the terminal G-quartet while surrounding adenine “propeller loops” create a constrained and highly complementary binding environment that suppresses torsional motion of the dye and thereby reduces nonradiative decay pathways. In the bound state, the benzothiazole and methylquinoline rings of TO1-biotin become nearly coplanar and rigidified against the quadruplex surface, dramatically increasing fluorescence quantum yield and lifetime. Importantly, the Mango II scaffold is highly preorganized even before ligand binding, contributing to both its thermostability and unusually high fluorophore affinity. Crystal structures further demonstrated that fluorescence efficiency correlates strongly with ligand–RNA shape complementarity and conformational homogeneity: mutations such as A22U reduce structural plasticity within the binding pocket, improve fluorophore packing, increase fluorescence

enhancement, and lengthen excited-state lifetimes by further restricting dye flexibility. Together, these findings establish Mango II as an RNA analog of protein fluorophore cages, where precise steric confinement and π -stacking interactions within a G-quadruplex architecture stabilize the emissive state and enable bright RNA imaging in living cells.



Spinach fluoresces because its G-quadruplex architecture locks DFHBI into a rigid, planar, hydrogen-bonded configuration that suppresses nonradiative relaxation and allows efficient photon emission essentially functioning as an RNA version of GFP's chromophore pocket.
PDB: 4TS0, PTS2 -- Crystal structure of the Spinach RNA



Fluorogenic RNA aptamer systems are valuable tools in cancer as they allow researchers to see where specific RNA molecules are located inside cells. For example, the Mango aptamer can be genetically attached to an RNA of interest, and when a dye such as SALAD1 binds to the aptamer, the RNA becomes fluorescent. This can help cancer researchers study whether an oncogenic transcript is concentrated in the nucleus, exported to the cytoplasm, or gathered in RNA granules that may affect tumor-cell survival.

Applications of aptamers bio-tools

Fluorescent RNA aptamers have important applications in studying disease pathophysiology; they allow researchers to monitor RNA activity, localization and abundance inside cells. In cancer research, these tools are especially useful because many disease mechanisms are controlled by RNA, including microRNAs, long noncoding RNAs and messenger RNAs that regulate oncogenes or tumor suppressor genes.

In 2021, Dou and colleagues developed a genetically encoded dual-color light-up RNA sensor for ratiometric imaging of microRNA. Their RNA-based fluorescent sensor allowed researchers to monitor microRNA activity inside cells (Dou et al., 2021). This is relevant to cancer research because microRNAs often regulate tumor suppressor genes and oncogenes. Similar aptamer-based reporters could be used to study whether a cancer-associated microRNA reduces the amount of a target RNA in the cytoplasm, redirects that RNA to processing bodies, or changes its localization after treatment with a targeted therapy. This would help researchers connect microRNA activity to cancer processes such as altered gene expression, uncontrolled proliferation, invasion and drug resistance.

Fluorescent aptamers are also useful in neurodegenerative disease research, where abnormal RNA repeats or RNA aggregates can disrupt normal cell function. In 2013, Strack, Disney and Jaffrey used the fluorescent RNA aptamer Spinach2 to study toxic repeat-containing RNA in mammalian cell-culture models. They engineered a reporter RNA containing expanded CGG repeats, which are associated with fragile X-associated tremor/ataxia syndrome, and attached Spinach2 to the repeat-containing RNA. When Spinach2 bound its fluorophore, the tagged RNA became fluorescent, allowing researchers to observe CGG-repeat RNA foci inside living cells and monitor how those foci changed after drug treatment (Strack, Disney and Jaffrey, 2013).

Spinach2-based microRNA assays also have value for studying cancer pathophysiology. In 2016, Aw, Tang, Teo and Cohen developed Pandan, a Spinach2-based fluorescent microRNA sensor, at research institutions in Singapore and Denmark. They redesigned the Spinach2 RNA aptamer so that it regained fluorescence only when it bound a specific target microRNA. This allowed the researchers to detect microRNAs using a visible fluorescence signal rather than relying only on indirect amplification-based methods (Aw et al., 2016).

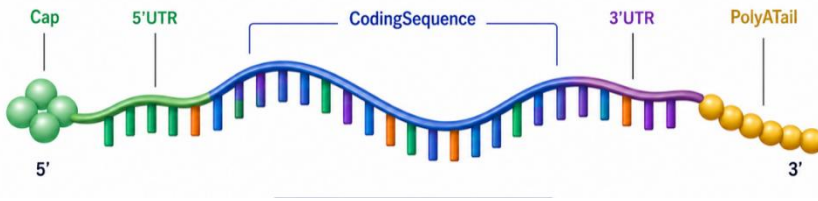
The value of this type of assay is that microRNAs are not simply biomarkers; they can actively shape disease mechanisms. In cancer, some microRNAs suppress tumor suppressor genes, while others reduce oncogene expression. For example, if a microRNA such as miR-21 is overactive in a cancer model, researchers could use a related fluorescent aptamer sensor to measure its activity and link that activity to downstream effects such as reduced tumor suppressor expression, increased proliferation, invasion or resistance to therapy.

These assays can also show how microRNA activity changes under different disease conditions. Researchers could compare fluorescence signals between healthy cells, early-stage cancer cells and aggressive cancer cells to determine whether a specific microRNA becomes more active as disease progresses. They could also treat cancer cells with chemotherapy, targeted therapy or a microRNA inhibitor and then measure whether the fluorescent signal decreases. This would help show whether treatment changes the RNA regulatory pathways that drive disease.

In 2024, Spence, Waldron and colleagues at the University of Aberdeen used a TDP-43 RNA aptamer, called TDP-43APT, to detect pathological TDP-43 in formalin-fixed paraffin-embedded human ALS tissue. The researchers applied the aptamer to post-mortem tissue from a deeply phenotyped ALS cohort and used it alongside single-molecule in situ hybridization to measure STMN2 cryptic splicing, a marker of TDP-43 loss of function. This allowed them to compare where abnormal TDP-43 appeared in cells with where RNA-splicing defects appeared in the same disease context. Their study showed that TDP-43APT detected pathological TDP-43 changes that standard antibody staining did not detect. In particular, the aptamer revealed abnormal nuclear TDP-43 pathology before obvious cytoplasmic TDP-43 accumulation. The researchers also found that this early nuclear pathology coincided with STMN2 cryptic splicing, suggesting that TDP-43 dysfunction begins before the later cytoplasmic aggregates usually associated with ALS pathology become prominent. This finding is valuable for understanding ALS pathophysiology because it changes how researchers think about the timing of TDP-43 disease mechanisms. Traditional pathology often focuses on cytoplasmic TDP-43 inclusions, but

Spence and colleagues showed that earlier nuclear changes may already reflect TDP-43 loss of function. Their work highlights that aptamer-based staining can reveal subtle early aggregation events, link those events to RNA-processing defects, and help identify disease changes that may occur before clinical symptoms or advanced pathology are obvious (Spence and Waldron et al., 2024).

mRNA Vaccines



For much of the 1960s and early 1970s, messenger RNA (mRNA) was viewed primarily as a passive intermediary between DNA and protein synthesis, with research focused largely on coding sequences while non-coding regions were considered biologically insignificant. Early studies of bacterial translation established foundational principles of translational initiation, including identification of AUG as the primary start codon and discovery of the Shine–Dalgarno sequence, a ribosome-binding motif that positioned prokaryotic ribosomes at initiation sites (Shine and Dalgarno, 1974). However, the absence of comparable motifs in eukaryotic mRNAs, together with the presence of long and structurally complex untranslated regions, suggested that eukaryotes used fundamentally different mechanisms to regulate translation. Subsequent discoveries progressively redefined mRNA as a highly organized regulatory molecule, beginning at the 5' end with identification of the 7-methylguanosine cap and the ribosomal scanning model proposed through the pioneering work of Marilyn Kozak, which established the importance of sequence context surrounding the AUG initiation codon (Kozak, 1984; Kozak, 1987). Later investigations extended this framework across the full length of the transcript, revealing regulatory roles for 5' untranslated regions (5' UTRs), coding sequence optimization, RNA secondary structure, modified nucleosides, poly(A) tails, and 3' untranslated regions (3' UTRs) in controlling mRNA stability, localization, and translational efficiency. Together, these discoveries transformed understanding of gene expression from a simple linear flow of information into a coordinated system of post-transcriptional regulation extending from the 5' to the 3' end of the molecule. These discoveries provided the technology stack for synthetic mRNA engineering modern mRNA therapeutics, including vaccines and programmable gene-delivery platforms.

The 5' Cap

In the 1970s, Aaron Shatkin and colleagues discovered that eukaryotic messenger RNAs contain an unusual, methylated guanosine linked to the 5' end through a 5'–5' triphosphate bridge (Shatkin, 1976).

They subsequently identified this structure as 7-methylguanosine (m7G) and demonstrated that the cap plays essential roles in mRNA stability and translational efficiency.

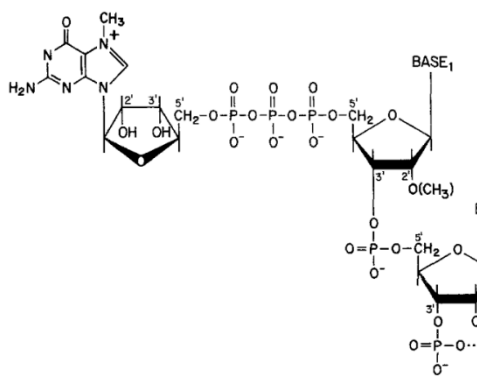


Figure 1. Structure of 5' Cap

Shatkin, A.J. (1976) 'Capping of eucaryotic mRNAs', *Cell*, 9(4 Pt 2), pp. 645–653. doi:10.1016/0092-8674(76)90128-8.

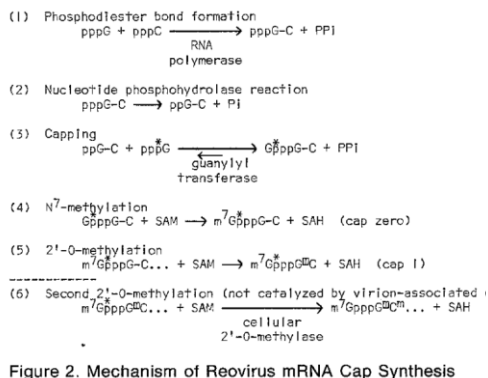
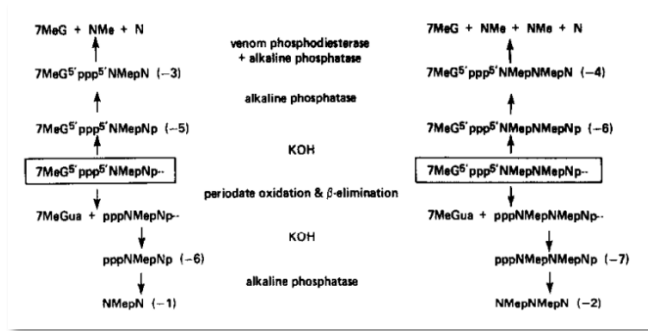


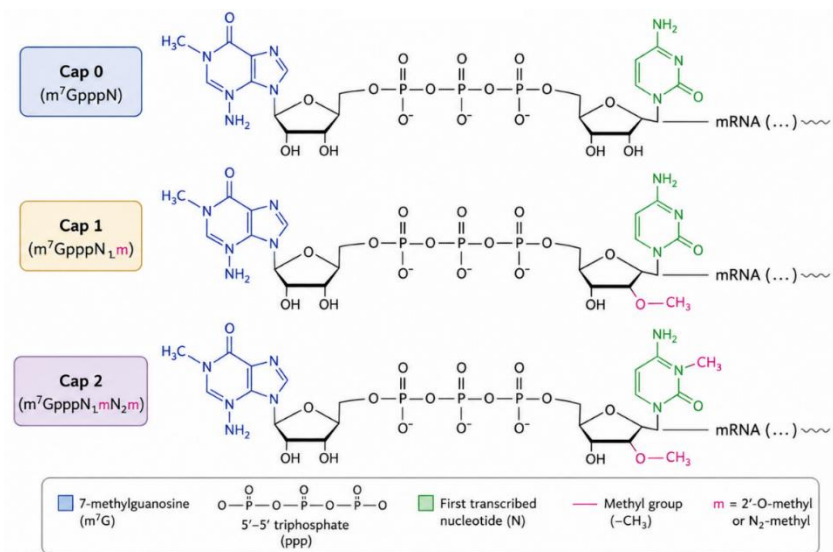
Figure 2. Mechanism of Reovirus mRNA Cap Synthesis

Over the following decades, researchers established that the 5' cap functions not only as a structural modification, but as a regulatory element coordinating transcription, RNA processing, nuclear export, translation, and RNA degradation (Furuichi and Shatkin, 2000; Topisirovic et al., 2011); the cap is a structurally distinct feature of eukaryotic mRNA and suggested that it played specialized roles. During the following decades, researchers discovered that eukaryotes frequently modify the first and second transcribed nucleotides through ribose 2'-O methylation.



IN THIS EXPERIMENT, RESEARCHERS CAREFULLY BROKE APART MESSENGER RNA FROM HUMAN HELa CELLS AND TRACKED THE CHEMICAL PIECES AT THE VERY FRONT END OF THE MOLECULE. BY USING A SERIES OF CHEMICAL TREATMENTS AND SEPARATION METHODS, THEY DISCOVERED THAT THE RNA CARRIED A SPECIAL MODIFIED "CAP" STRUCTURE.

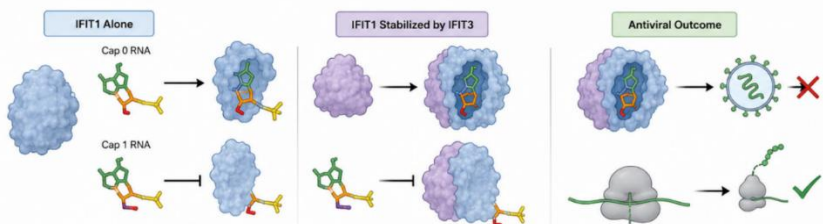
In 1975, Cha-Mer Wei, Alan Gershowitz, and Bernard Moss at the National Institute of Allergy and Infectious Diseases discovered that messenger RNA in human cells carries a previously unknown methylated structure at its 5' end; this was later termed "Cap 1". Using radiolabeled HeLa cell poly(A)+ mRNA together with enzymatic digestion, DEAE-cellulose chromatography, and periodate oxidation experiments, researchers identified a blocked 5' terminal sequence containing 7-methylguanosine linked through multiple phosphates to 2'-O-methylated nucleotides. Their work provided the first evidence that eukaryotic mRNAs possess a specialized methylated cap structure that likely plays an important regulatory role in mRNA function and stability.



Decades later, biochemical and virus studies clarified the immunological significance of 2'-O-methylation within the mRNA cap structure. Experiments comparing Cap 0 and Cap 1 RNAs demonstrated that interferon-induced proteins such as IFIT1 selectively bind and suppress translation of RNAs lacking 2'-O-methylation, whereas properly methylated Cap 1 RNAs evade this restriction and undergo efficient translation (Daffis et al., 2010; Habjan et al., 2013). Viral genetic studies further showed that viruses lacking functional 2'-O-methyltransferase activity, and therefore unable to generate proper Cap 1 structures, were highly sensitive to innate immune responses, confirming that Cap 1 methylation functions as a molecular signature of self RNA (Züst et al., 2011). Together, these studies established that 2'-O-methylation of the mRNA cap constitutes a critical mechanism by which eukaryotic cells distinguish self from non-self RNA.

Building on these findings, researchers at the Institute of Immunobiology, Kanton Hospital St. Gallen, Switzerland investigated how ribose 2'-O-methylation regulates innate immune recognition during coronavirus infection. Using genetically engineered human and mouse coronaviruses lacking functional 2'-O-methyltransferase enzymes, investigators demonstrated that the cytoplasmic RNA sensor Mda5 specifically recognizes viral RNAs lacking proper 2'-O-methylation. By infecting macrophages with methylation-deficient coronaviruses, they observed activation of interferon regulatory factor 3 (IRF3), elevated IFN- β production, and enhanced antiviral signaling pathways. In contrast, macrophages deficient in Mda5 failed to mount these interferon responses, confirming that Mda5 functions as a key sensor of improperly methylated viral RNA (Züst et al., 2011).

These findings were further validated in mouse models of coronavirus infection. When they infected wild-type mice with methylation-deficient coronaviruses, the viruses replicated poorly and exhibited markedly reduced pathogenicity. However, viral replication was restored in mice lacking interferon receptors or the RNA sensors Mda5 and TLR7. These experiments demonstrated that Cap 1 methylation enables coronaviruses to evade innate immune recognition and establish productive infection within the host (Züst et al., 2011).



Several groups determined the structural basis of IFIT-mediated RNA discrimination through structure-based studies. Using high-resolution crystal structures of human IFIT1 bound to capped viral RNA, researchers demonstrated that IFIT1 forms a binding pocket that preferentially accommodates Cap 0 RNAs lacking ribose 2'-O-methylation, whereas Cap 1 methylation sterically interferes with stable IFIT1 engagement. They further showed that IFIT3 stabilizes IFIT1 and enhances its selective recognition of unmethylated viral RNAs, thereby strengthening antiviral restriction of viruses deficient in 2'-O-methyltransferase activity (Johnson et al., 2018). Together, these structural and biochemical studies provided direct mechanistic validation that 2'-O-methylation functions as a molecular signature of self RNA by preventing IFIT-mediated recognition and translational suppression of properly methylated host transcripts.

In 2025 Rahul Singh and collaborators in the laboratory of structural immunologist Yorgo Modis at the MRC Laboratory of Molecular Biology and the University of Cambridge in the United Kingdom investigated the molecular basis by which MDA5 distinguishes self from non-self RNA. Using cryo-electron microscopy, ATPase assays, RNA filament reconstruction, they demonstrated that MDA5 employs ATP-dependent proofreading and cooperative filament formation along double-stranded RNA to discriminate endogenous cellular RNA from viral RNA. The investigators further showed that disease-associated MDA5 variants altered RNA binding, ATPase activity, and interferon signaling, thereby directly linking defective RNA recognition to dysregulated antiviral immunity and susceptibility to autoimmune disease (Singh et al., 2025). Together, these findings reinforced the concept that innate immune sensors rely on highly precise structural recognition mechanisms to identify aberrant viral RNA while avoiding inappropriate immune responses against host RNA.

Ribose 2'-O-methylation is a critical molecular signature of self RNA. By mimicking host mRNA cap structures, viruses and synthetic therapeutic mRNA evade innate immune detection and escape interferon-mediated antiviral restriction.

Translation Initiation

The identification of AUG as the universal initiation codon emerged from early efforts to decipher the genetic code and understand how ribosomes establish the correct reading frame during protein synthesis. In 1966, T. A. Sundararajan and Robert E. Thach, working in the Department of Chemistry at Harvard University under the broader influence of the pioneering molecular biologist Paul Doty, demonstrated that the triplet AUG